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CMSC 207 CRN 32950

CMSC 207: Chapter 5 Homework

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1. Use Mathematical Induction to prove that for all integers $n \geq 3$,

$$2 \cdot 3 + 3 \cdot 4 + \ldots + (n-1) \cdot n = \frac{(n-2)\left(n^2 + 2n + 3\right)}{3}.$$

Let $P(n)$ be the statement:

$$2 \cdot 3 + 3 \cdot 4 + \ldots + (n-1) \cdot n = \frac{(n-2)\left(n^2 + 2n + 3\right)}{3}.$$

Base case: Let $n = 3$. Then:

$$\begin{aligned} 2 \cdot 3 &= \frac{(3-2)\left(3^2 + 2(3) + 3\right)}{3} \\ 6 &= \frac{1(9+6+3)}{3} \\ 6 &= \frac{1(18)}{3} \\ 6 &= 6. \end{aligned}$$

Since the left hand side equals the right hand side, $P(3)$ is true.

Inductive hypothesis: Assume $P(k)$ is true for some integer $k \geq 3$. That is, assume:

$$2 \cdot 3 + 3 \cdot 4 + \ldots + (k-1) \cdot k = \frac{(k-2)\left(k^2 + 2k + 3\right)}{3}.$$

Inductive step: We want to show that $P(k+1)$ is true, i.e. we want to show:

$$2 \cdot 3 + 3 \cdot 4 + \ldots + (k-1) \cdot k + k \cdot (k+1) = \frac{((k+1)-2)\left((k+1)^2 + 2(k+1) + 3\right)}{3}.$$

Starting with the left hand side of $P(k+1)$, we can use the inductive hypothesis to rewrite it as:

$$\begin{aligned} 2 \cdot 3 + 3 \cdot 4 + \ldots + (k-1) \cdot k + k \cdot (k+1) &= \frac{(k-2)\left(k^2 + 2k + 3\right)}{3} + k \cdot (k+1) \\ &= \frac{(k-2)\left(k^2 + 2k + 3\right)}{3} + \frac{3k(k+1)}{3} \\ &= \frac{\left((k-2)\left(k^2 + 2k + 3\right)\right) + (3k(k+1))}{3}. \end{aligned}$$

Expanding the numerator, we have:

$$\begin{aligned}\left((k-2)\left(k^2+2k+3\right)\right)+(3k(k+1)) &= \left(k^3+2k^2+3k\right)-\left(2k^2+4k+6\right)+\left(3k^2+3k\right)\\ &= k^3+3k^2+2k-6.\end{aligned}$$

Thus, the left hand side of $P(k+1)$ can be rewritten as:

$$2\cdot 3+3\cdot 4+\ldots+(k-1)\cdot k+k\cdot(k+1)=\frac{k^3+3k^2+2k-6}{3}.$$

Now, we can simplify the right hand side of $P(k+1)$:

$$\begin{aligned}\frac{((k+1)-2)\left((k+1)^2+2(k+1)+3\right)}{3} &= \frac{(k-1)\left(k^2+4k+6\right)}{3}\\ &= \frac{k^3+3k^2+2k-6}{3}.\end{aligned}$$

Since the left hand side and right hand side of $P(k+1)$ are equal, $P(k+1)$ is true.

Since assuming the truth of $P(k)$ for some integer $k \geq 3$ leads to the truth of $P(k+1)$, $P(n)$ is true for all integers $n \geq 3$ by mathematical induction. ∎

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2. Use Mathematical Induction to prove that for all integers $n \geq 5$,

$$1 + 4n < 2^n.$$

Let $P(n)$ be the statement:

$$1 + 4n < 2^n.$$

Base case: Let $n = 5$. Then the inequality can be rewritten as:

$$\begin{aligned} 1 + 4(5) &< 2^5 \\ 21 &< 32. \end{aligned}$$

Since $21 < 32$, $P(5)$ is true.

Inductive hypothesis: Assume $P(k)$ is true for some integer $k \geq 5$. That is, assume:

$$1 + 4k < 2^k.$$

Inductive step: We want to show that $P(k+1)$ is true, i.e. we want to show:

$$1 + 4(k+1) < 2^{k+1}.$$

Starting from the inductive hypothesis:

$$\begin{aligned} 1 + 4k &< 2^k && \text{assumed by inductive hypothesis} \\ 1 + 4k + 4 &< 2^k + 4 && \text{by adding 4 to both sides} \\ 1 + 4(k+1) &< 2^k + 4 && \text{by algebra of lhs} \\ 1 + 4(k+1) &< 2^k + 2^k && \text{since } 4 < 2^k \text{ for all } k \geq 5 \\ 1 + 4(k+1) &< 2^{k+1} && \text{by algebra of rhs} \end{aligned}$$

Since assuming the truth of $P(k)$ for $k \geq 5$ leads to the truth of $P(k+1)$, $P(n)$ is true for all integers $n \geq 5$ by mathematical induction. ∎